The Closure Geometry Reaches ζ_K, Not ζ: a Certified Negative on the 600-Cell Bridge

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Abstract

The icosian / 600-cell closure geometry realizes, over $K = \mathbb{Q}(\sqrt{5})$, the Dedekind-type L-function $\zeta_K(s)\,\zeta_K(s-1)$; this is established in prior work of the programme and recalled here. We ask the natural follow-on question—whether the *same* geometry realizes the Riemann zeta function ζ itself—and answer it in the negative. Using a reproducible, gate-first verification engine, we certify that the proposed "substrate $\to \zeta$" correspondence **fails** on several independent diagnostics (zero positions, nearest-neighbour spacing, on-line density, and degree), confirmed by two largely-disjoint routes (a spacing route and a counting route). The conclusion is a mapped boundary: the closure geometry's arithmetic reach is exactly $\zeta_K(s)\zeta_K(s-1)$—which contains ζ only as a factor—and not ζ. **No claim is made about the Riemann Hypothesis**, and no positivity, operator, arithmetic-surface, or physical/cosmological claim is made or implied. The engine issues verification certificates only; it defines and issues no proof certificate.

1 Introduction and scope

The 600-cell is the regular 4-polytope whose 120 vertices are the unit icosians, i.e. the binary icosahedral group $2I$; the icosian ring is a maximal order in a definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$, and its underlying lattice is a copy of E_8 [1]. Prior work of this programme [3] established, with out-of-sample numerical verification, that the icosian theta series realizes the rank-two product $\zeta_K(s)\zeta_K(s-1)$. This note takes that result as input (the *reach*) and settles a single follow-on question (the *limit*).

What is claimed. (i) A recollection of the verified correspondences—icosian $\to \zeta_K(s)\zeta_K(s-1)$, the rational 24-cell $\to \zeta(s)\zeta(s-1)$, the φ-shell $\to L(s,\chi_5)$ (§3); and (ii) a *certified negative*: the natural "substrate $\to \zeta$" correspondence does not hold, by a reproducible verification gate (§4).

What is not claimed. Nothing about the location of the zeros of ζ (the Riemann Hypothesis); no positivity, self-adjointness, operator, or arithmetic-surface statement; no connection to any physical or cosmological model. These boundaries are deliberate and load-bearing. The negative of §4 is a statement that a specific *construction* fails to realize ζ; it is not a statement about the zeros of ζ and implies nothing about RH.

2 Preliminaries

Let $K = \mathbb{Q}(\sqrt{5})$, $\varphi = \frac{1+\sqrt{5}}{2}$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$. The Dedekind zeta function factors classically as

$$\zeta_K(s) = \zeta(s)\,L(s,\chi_5), \tag{1}$$

with χ_5 the even real quadratic character mod 5 [2]; ζ has a simple pole at $s = 1$, and $L(\cdot, \chi_5)$ is entire. The icosian ring $\mathcal{I}$ is the $\mathbb{Z}[\varphi]$-order on the 120 unit icosians; its 24-element rational substructure is the ring of Hurwitz units (the 24-cell / $2T$, over $K = \mathbb{Q}$), and the remaining 96 vertices form the φ-shell. The coordinate Galois involution $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ fixes the 24-cell pointwise and permutes the φ-shell.

Lemma 1 (class number one)**.** *Let B be the definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$ ramified exactly at the two infinite places, and $\mathcal{I} \subset B$ the icosian maximal order. Then $\mathcal{I}$ has a single left-ideal class.*

Reference. $(B, \mathcal{I})$ is the maximal-order case with invariants $(n, d_F, D, N) = (2, 5, 1, 1)$ in the Kirschmer–Voight enumeration of definite quaternion orders of class number one over totally real fields [4, Table 8.2] (see also Vignéras [5]). This is the left-ideal class number entering the Siegel–Weil average used in [3]. □

3 The reach: verified correspondences

We recall the established positive correspondences; the first is the principal input, proved and out-of-sample verified in [3] and not reproved here.

Proposition 2 (recalled from [3])**.** *The Hilbert theta series of the icosian order over K (weight two, level $(D, N) = (1, 1)$, normalized to leading coefficient 1) satisfies $L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\,\zeta_K(s-1)$. By Siegel–Weil and Lemma 1 (single class), the theta series equals the corresponding Eisenstein series with no cuspidal contribution; the representation numbers were obtained by exact short-vector enumeration over $\mathbb{Z}[\varphi]$ and matched, out of sample, to the Dirichlet coefficients of $\zeta_K(s)\zeta_K(s-1)$.*

Certified Result 3. The rational 24-cell (Hurwitz) theta series realizes $\zeta(s)\,\zeta(s-1)$.

Proposition 4. *On the critical line $\mathrm{Re}(s) = \frac{1}{2}$, the zeros of $\zeta(s)\zeta(s-1)$ are exactly those nontrivial zeros of ζ that lie on that line; the factor $\zeta(s-1)$ contributes none of them.*

Proof. A nontrivial zero of $\zeta(s-1)$ sits at $s = 1 + \rho$ with $\mathrm{Re}(\rho) \in (0, 1)$, hence $\mathrm{Re}(s) \in (1, 2)$; the trivial zeros of $\zeta(s-1)$ are at $s = 1 - 2n$ $(n \geq 1)$. Neither meets $\mathrm{Re}(s) = \frac{1}{2}$, so the on-line zeros of the product are precisely the on-line zeros of $\zeta(s)$. This concerns *which* zeros lie on the line and asserts nothing about whether all nontrivial zeros of ζ do. □

Certified Result 5. The 96-vertex φ-shell (the Galois-sign component) realizes $L(s, \chi_5)$: its on-line zeros coincide, to tested precision, with those of $L(s, \chi_5)$; no coincidence with a Riemann zero was observed in the tested range.

Via (1), these are mutually consistent: the full icosian object carries $\zeta_K(s)\zeta_K(s-1) = \zeta(s)L(s, \chi_5)\zeta(s-1)L(s-1, \chi_5)$; its Galois-fixed part carries the ζ-factor (the 24-cell), its Galois-sign part the $L(s, \chi_5)$ factor (the φ-shell).

4 The limit: a verification gate and the negative

4.1 The gate

A proposed correspondence ("expression") supplies a parameter-free procedure that produces the on-line zeros of a claimed L-function *without taking the target as input.* The gate evaluates: **no-hardcoding** (the target is not an input), **provenance** (no constant declared fitted/tuned), **fingerprint** (nearest-neighbour spacing; fraction of normalized gaps below 0.3), **pointwise** (agreement of the first eight zeros, tolerance 0.2), **density** (on-line zeros per unit height), **rigidity** (number variance $\Sigma^2(\ell)$), and structural **degree** and **pole**. An expression is *verified* only if it passes all applicable checks; a hashed certificate is emitted either way. The engine is a *certifier*, not a search.

Honest reading of the gate. The **no-hardcoding** and **provenance** checks act on the expression's declared procedure and provenance string; they catch a *declared* fitted map but rely on honest declaration, not on program analysis. Accordingly the gate certifies the *consistency* of a declared correspondence against the named target's invariants; the geometry→arithmetic enumeration itself is the separate computation of [3], cited above, not re-performed by the gate.

4.2 The certified negative

Certified Result 6 (the $(O2)$ negative)**.** The proposed correspondence "substrate $\to \zeta$" is **rejected**. The substrate's on-line content is that of $\zeta_K(s)\zeta_K(s-1)$ (degree 4), which by (1) is $\zeta(s)L(s,\chi_5)\zeta(s-1)L(s-1,\chi_5)$; this contains ζ only as a factor. Against the target ζ (degree 1) the gate fails on: *degree* (4 vs 1); *fingerprint* (spacing fraction 0.077 for the superposition vs 0.000 for a single ζ-type spectrum); *pointwise* (maximum displacement over the first eight zeros ≈ 22.3; the first-zero displacement alone is ≈ 7.5); and *density* (see Result 7).

Certified Result 7 (independent counting route)**.** A route largely disjoint from the spacing diagnostics: since the substrate carries the union of the zeros of ζ and of $L(\chi_5)$ (and their $s-1$ shifts), its on-line density strictly exceeds that of ζ—empirically a ratio $\approx 4.7, 3.7, 3.1$ at heights $30, 40, 50$, tending to 2 asymptotically as each factor contributes $\sim \frac{1}{2\pi}\log\frac{T}{2\pi}$. A spectrum strictly denser than ζ's is not ζ's. (This route shares only the *density* diagnostic with Result 6; the spacing/fingerprint evidence is disjoint from it.)

4.3 Controls

Certified Result 8 (fitted-map control)**.** A correspondence whose provenance *declares* a least-squares fit to the zeros is rejected on **provenance** alone, independent of how well its output matches. This reproduces, mechanically, a fitted-map trap caught in a prior internal audit of this programme.

Certified Result 9 (class-vs-individual control)**.** A generic GUE operator (random Hermitian, no arithmetic) *passes* the **fingerprint** check—it has the right universality class—but *fails* **pointwise** (it is not the right individual spectrum), and in the saved atlas also fails density and rigidity. We rest this control on the pointwise failure: GUE-class membership is shown to be insufficient to be ζ. (The density/rigidity numbers here depend on a window-scaling convention and are not advanced as intrinsic individual discriminators.)

expression	verdict	failing checks
24-cell $\to \zeta(s)\zeta(s-1)$	VERIFIED	—
icosian $\to \zeta_K(s)\zeta_K(s-1)$	VERIFIED	—
φ-shell $\to L(s,\chi_5)$	VERIFIED	—
substrate $\to \zeta$ (bridge)	REJECTED	degree, fingerprint, pointwise, density
fitted map $\to \zeta$	REJECTED	provenance
generic GUE $\to \zeta$	REJECTED	pointwise (class $\neq$ individual)

5 Reproducibility and scope

The gate, certificates (content-hashed), and the atlas accompany this note in `vfd_math_engine/`. We claim only what the certificates state: which correspondences pass the gate and which do not. The geometry→arithmetic enumeration underlying Proposition 2 is the separate, out-of-sample computation of [3].

The net statement is a *boundary*: the closure geometry realizes $\zeta_K(s)\zeta_K(s-1)$ and provably (within the gate's scope) not ζ. We make no claim about the Riemann Hypothesis. No proof certificate is defined or issued.

References

[1] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Springer.

[2] J. Neukirch, *Algebraic Number Theory*, Springer.

[3] VFD programme, *The Icosian Triad on* V_{600} (and companion computations), `icosian-triad-v600` release; establishes $L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$ with out-of-sample verification.

[4] M. Kirschmer, J. Voight, *Algorithmic enumeration of ideal classes for quaternion orders*, SIAM J. Comput. **39** (2010), Table 8.2.

[5] M.-F. Vignéras, *Arithmétique des algèbres de quaternions*, Lecture Notes in Math. **800**, Springer (1980).

[6] C. L. Siegel, *Über die analytische Theorie der quadratischen Formen*, Ann. of Math. (1935).

[7] A. Weil, *Sur la formule de Siegel...*, Acta Math. (1965).